Building Airflow ETL Pipeline

Currency Exchange Rates to MongoDB

Name	M. Abdul Raheem
Roll No.	23L-0588
Section	BCS-7A
Repository	github.com/abdul-raheem-fast/currency-exchange-airflow-etl

1. Objective

The project implements a daily Apache Airflow ETL pipeline that extracts EUR exchange rates from the Frankfurter API, transforms the response into the required document schema, and loads the records into MongoDB.

2. Pipeline Implementation

Task	Implementation
Extract	Calls Frankfurter API for USD, GBP, and JPY; pushes raw_rates through named XCom.
Transform	Creates one document per currency with base, target, rate, date, source, and retrieval timestamp.
Load	Reads transformed_rates through named XCom and inserts documents into currency_db.exchange_rates

Task dependency: extract >> transform >> load

3. Transformed Data Schema

```
{"base_currency": "EUR", "target_currency": "USD", "exchange_rate": 1.15,
"rate_date": "2026-11-04", "source": "Frankfurter API", "retrieved_at":
"2026-11-05T07:15:00"}
```

4. Infrastructure

The Astro Airflow project includes MongoDB 7.0 and Mongo Express Docker services. MongoDB uses the persistent mongo_data volume. Both services are connected to the Airflow network, and Mongo Express is available on port 8081. Docker image metadata labels include maintainer, description, and version.

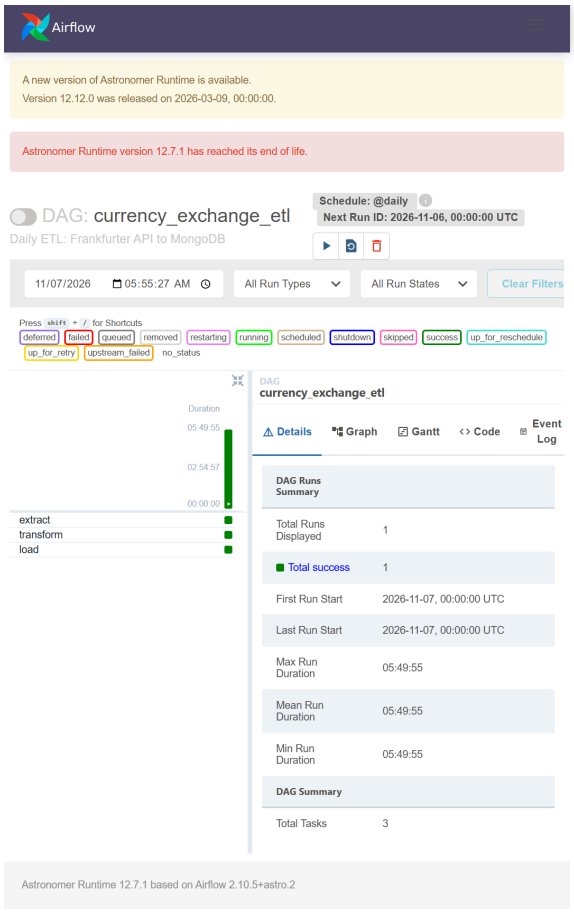
5. Requirements Checklist

Requirement	Status
Frankfurter API extraction	Completed
Three Airflow tasks	Completed
Named XCom transfer	Completed
Transformation schema	Completed
MongoDB load	Completed
Mongo Express service	Completed

Docker networking and persistent storage	Completed
GitHub repository	Completed

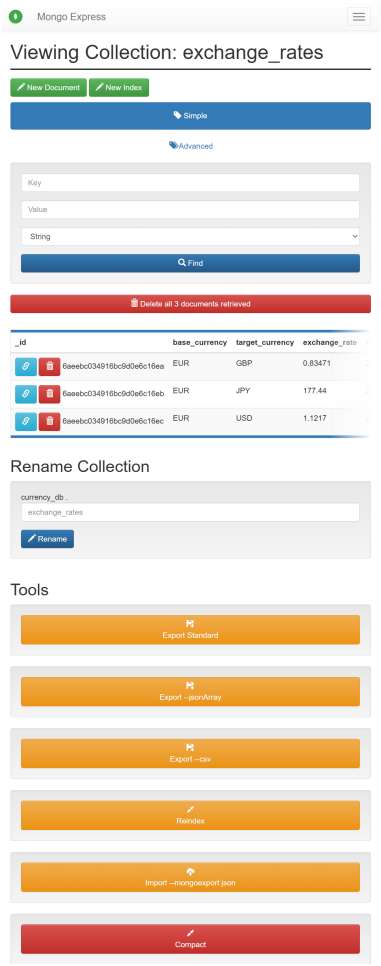
6. Runtime Evidence: Airflow

The Airflow grid confirms one successful DAG run and three successful tasks.



7. Runtime Evidence: Mongo Express

Mongo Express confirms that the exchange_rates collection contains the transformed EUR to GBP, JPY, and USD documents.



8. Submission Items

Submit this PDF report, the GitHub repository link, and the project source code. The repository contains the DAG, Docker configuration, tests, README, and report source.